L05: Learning in games

Algorithms



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How do we construct a regret minimizer?

Two philosophies

1. Prioritize high-regret actions

Quite natural: use the past to inform what to do next.

FTL · RM · MWU · (Generally: FTRL)

↑ **We start here**

2. Improve locally

Take a local improvement (gradient) step.

OGD · MWU · (Generally: OMD)

Let's track regrets

At every time t, the *instantaneous regret* accumulated against each action is

$$\delta_a^{(t)} := g_a^{(t)} - \langle \boldsymbol{g}^{(t)}, \boldsymbol{x}^{(t)} \rangle$$

In vector form:

$$\boxed{\boldsymbol{\delta}^{(t)} := \boldsymbol{g}^{(t)} - \langle \boldsymbol{g}^{(t)}, \boldsymbol{x}^{(t)} \rangle \mathbf{1}}$$

Hence, the vector of regrets against each action is

$$\boxed{\boldsymbol{r}^{(t)} := \sum_{\tau=1}^{t} \boldsymbol{\delta}^{(\tau)}}, \qquad \text{and} \qquad \boxed{\mathrm{Reg}^{(T)} = \max_a r_a^{(T)}}$$

Follow-the-leader

Idea #1: Pick the action with *highest regret*

Follow-the-leader

Play the action we most wish we had used in the past.

All mass on the action with highest regret

$$\boldsymbol{x}^{(t+1)} = \boldsymbol{e}_{\arg\max_a r_a^{(t)}}$$

Equivalently, maximize cumulative utility:

$$\arg\max_a r_a^{(t)} = \arg\max_a \sum_{\tau=1}^{t} g_a^{(\tau)}$$

What could go wrong?

Follow-the-leader

t	1	2	3	4	5	6
FTL plays	a_1	a_2	a_1	a_2	a_1	a_2
$g^{(t)}$	$(0, 0.5)$	$(1, 0)$	$(0, 1)$	$(1, 0)$	$(0, 1)$	$(1, 0)$
FTL total	0	0	0	0	0	0
Best fixed total	0.5	1	1.5	2	2.5	3

Regret: 3

The leader can always be one round behind: $\mathrm{Reg}^{(T)}$ could be $\Omega(T)$ in general)

Intuitive issue: the algorithm "*jumps too much*". It is "not smooth"...

Not all hope is lost!

Interesting result: If both players in a *two-player zero-sum game* use FTL, we obtain *fictitious play*.

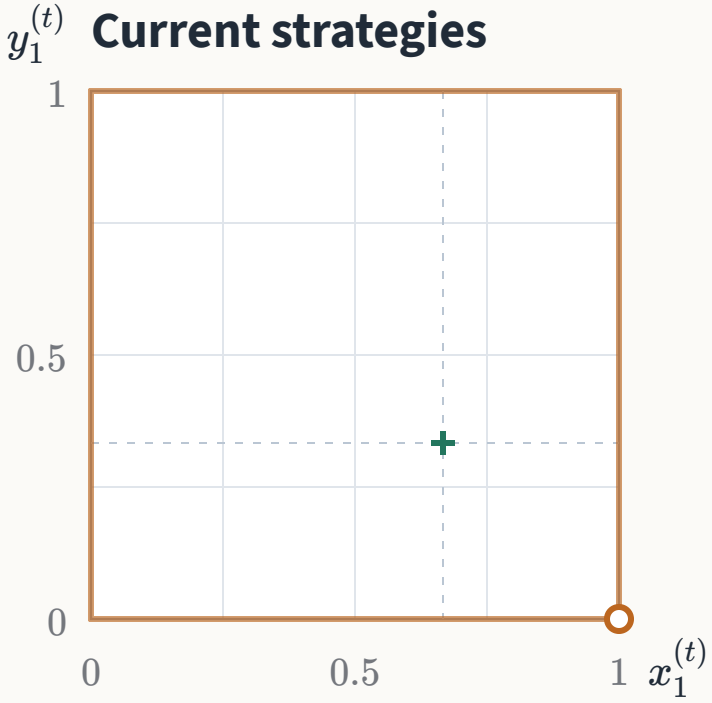
> **Julia Robinson (1951)**
> In every finite two-player zero-sum game, the *average strategies* approach the set of Nash equilibria.
> *(An iterative method of solving a game, 1950.)*

Convergence can be very slow. But this shows that if everyone is learning, FTL has sublinear regret...

Fictitious play in a zero-sum game

Current play keeps switching; averages approach Nash

Each player best responds to the opponent's average strategy so far.



$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$u_1 = \boldsymbol{x}^\top U \boldsymbol{y}$

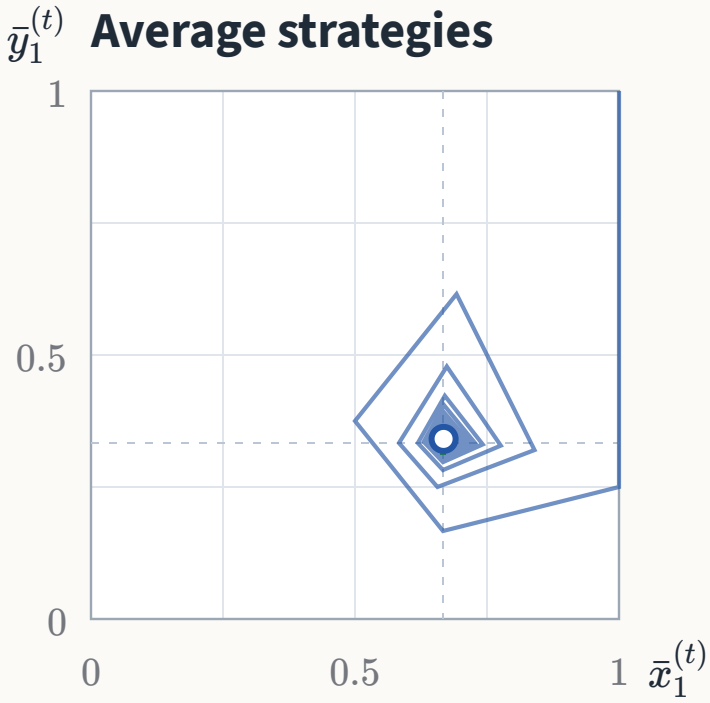
$u_2 = -u_1$

Pure best responses

$(x_1, y_1) = (1, 0)$

Equilibrium gap

1.000



+ Nash equilibrium

$(x_1^*, y_1^*) = (\frac{2}{3}, \frac{1}{3})$

Running average

$(0.668, 0.341)$

Equilibrium gap

0.009

Simultaneous updates · Ties: choose the first action

How do we prove the no-regret property?

Two approaches

1. Against arbitrary adversaries

Seek to prove that the property holds no matter what the environment does

Pros: no need to think about the world, robust
Cons: higher bar, more conservative/pessimistic

↑ ***Online learning*** **approach**

2. Within a more benign environment

If everyone is learning, the dynamics are somewhat smooth. Full adversarialness is not unleashed

Pros: better results *are* possible (eg, *fictitious play*)
Cons: analysis is harder, makes assumptions about the environment

Regret matching

Idea #2: Pick actions *proportional** to their regret

* Wait a second: regret can be negative. So let's first threshold at zero...

Regret matching

Second simplest idea: allocate mass proportional to regret

Use the positive part: $[r_a]^+ := \max\{r_a, 0\}$.

Play in proportion to positive regret

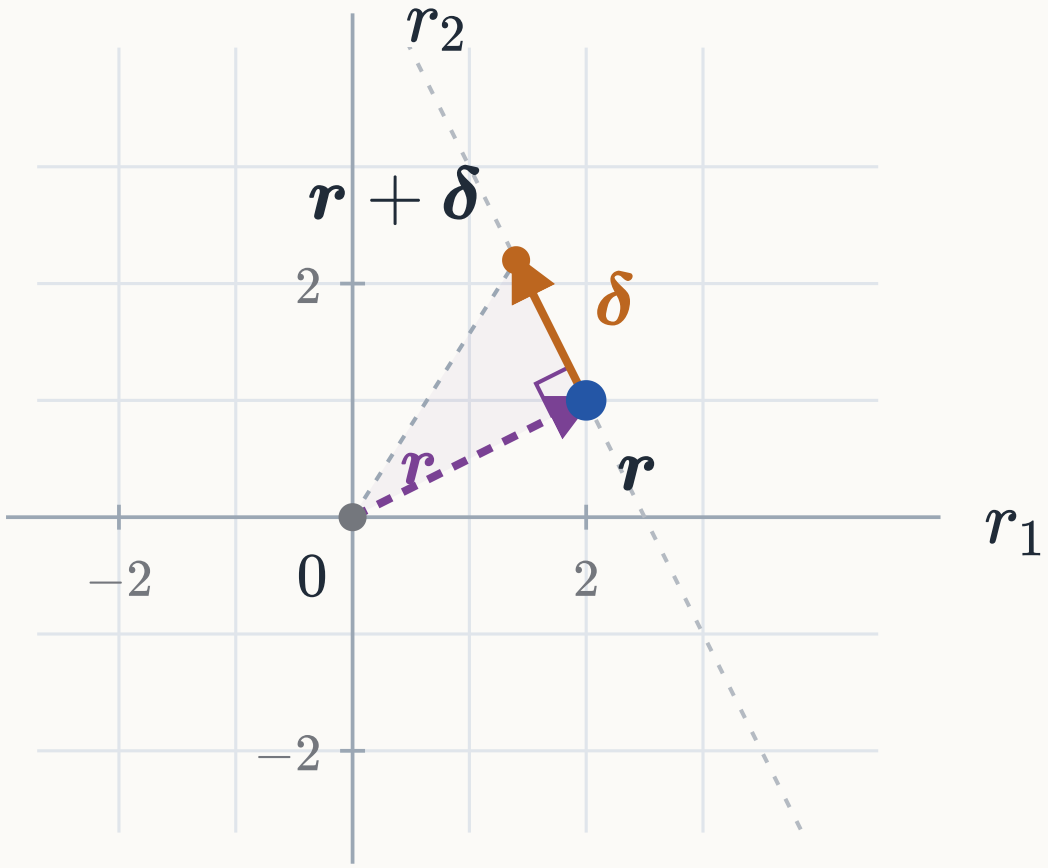
$$\boldsymbol{x}^{(t+1)} = \frac{\left[\boldsymbol{r}^{(t)}\right]^+}{\left\|\left[\boldsymbol{r}^{(t)}\right]^+\right\|_1}$$

If all regrets are nonpositive, use any distribution.

$\boldsymbol{r} = (2, -1, 1)$ $\qquad$ $\boldsymbol{x} = ?$

$$\boldsymbol{x} = (2/3, 0, 1/3)$$

The geometry of regret matching



First, a simplifying assumption

Suppose the current $\boldsymbol{r} \geq \mathbf{0}$.

The new regret increment

$$\boldsymbol{\delta} = \boldsymbol{g} - \langle \boldsymbol{g}, \boldsymbol{x} \rangle \mathbf{1}$$

Just Pythagoras

$\langle \boldsymbol{x}, \boldsymbol{\delta} \rangle = 0$ for every strategy.

RM chooses $\boldsymbol{x} \propto \boldsymbol{r}$.

$$\langle \boldsymbol{r}, \boldsymbol{\delta} \rangle = 0$$

$$\|\boldsymbol{r} + \boldsymbol{\delta}\|_2^2 = \|\boldsymbol{r}\|_2^2 + \|\boldsymbol{\delta}\|_2^2$$

$\boldsymbol{x} = (0.67,\ 0.33)$

The general case: regrets can be negative

Now drop the assumption. Track the squared norm of the positive part: $\|[\boldsymbol{r}]^+\|_2^2$

1. Orthogonality: $\langle [\boldsymbol{r}]^+, \boldsymbol{\delta} \rangle = \|[\boldsymbol{r}]^+\|_1 \langle \boldsymbol{x}, \boldsymbol{\delta} \rangle = 0.$

2. Inequality + Pythagoras:

$$\begin{aligned} \|[\boldsymbol{r} + \boldsymbol{\delta}]^+\|_2^2 &\leq \|[\boldsymbol{r}]^+ + \boldsymbol{\delta}\|_2^2 \\ &= \|[\boldsymbol{r}]^+\|_2^2 + \underbrace{2\langle [\boldsymbol{r}]^+, \boldsymbol{\delta} \rangle}_{=0} + \|\boldsymbol{\delta}\|_2^2 \end{aligned}$$

Hence,

$$\|[\boldsymbol{r}^{(T)}]^+\|_2^2 \leq \sum_{t=1}^{T} \|\boldsymbol{\delta}^{(t)}\|_2^2$$

Right-hand side scales linearly in T, which implies regret bounded by $\approx \sqrt{T}$...

Regret matching: regret bound

Assuming $\|\boldsymbol{g}^{(t)}\|_\infty \leq 1$ at all times t,

$$\begin{aligned}\|\boldsymbol{\delta}^{(t)}\|_2 &= \|\boldsymbol{g}^{(t)} - \langle \boldsymbol{g}^{(t)}, \boldsymbol{x}^{(t)} \rangle \mathbf{1}\|_2 \\ &\leq \|\boldsymbol{g}^{(t)}\|_2 + |\langle \boldsymbol{g}^{(t)}, \boldsymbol{x}^{(t)} \rangle| \cdot \|\mathbf{1}\|_2 \\ &\leq \sqrt{|A|} + \sqrt{|A|} = 2\sqrt{|A|}\end{aligned}$$

Hence

$$\mathrm{Reg}^{(T)} = \max_a r_a^{(T)} \leq \sqrt{\|\boldsymbol{r}^{(T)}\|_2^2} \leq \sqrt{T \cdot 4|A|} = 2\sqrt{T|A|}.$$

Great news! Sublinear regret with no stepsize to tune

$$\boxed{\mathrm{Reg}^{(T)} \leq 2\sqrt{T|A|}}$$

Against *any* sequence of bounded utility vectors. No stepsize to tune!

Trivia: Poker breakthroughs used (variants) of this

Multiplicative weights update

Idea #3: Avoid instability by *smoothing* the maximum

Smoothing follow-the-leader

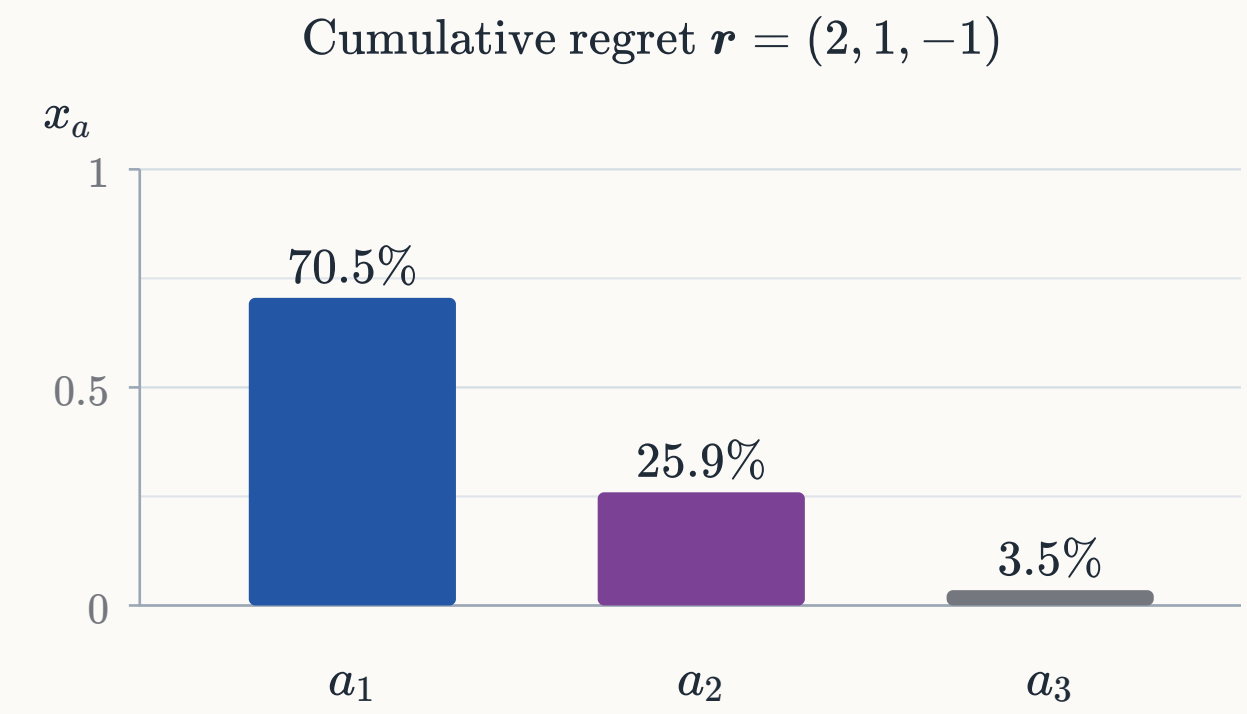
$$x_a = \frac{\exp(\eta r_a)}{\sum_b \exp(\eta r_b)}$$

$$\eta \downarrow 0 : \text{ uniform}$$

$$\eta \uparrow \infty : \text{ leader}$$

The stepsize controls sensitivity

Small stepsizes temper the influence of past scores. Large stepsizes approach the hard leader.



Small η: nearly uniform. Large η: nearly follow-the-leader.

MWU: regret bound

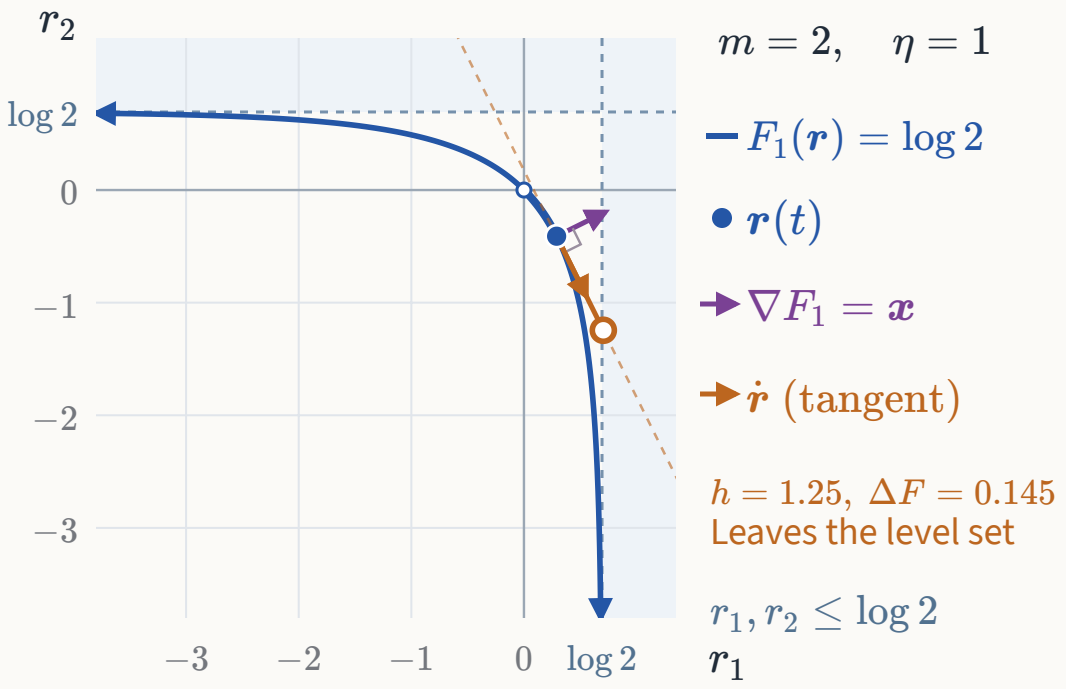
$$\boxed{\mathrm{Reg}^{(T)} \leq \frac{\log |A|}{\eta} + \eta \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_{\infty}^{2}}$$

Hence, by setting $\eta = \sqrt{\log |A| / T}$,

$$\mathrm{Reg}^{(T)} \leq 2\sqrt{T \log |A|}$$

Compared to RM: Logarithmic dependence on the number of actions, but one hyperparameter (η)

Continuous-time MWU



Softmax is the gradient of log-sum-exp

$$F_\eta(\boldsymbol{r}) := \tfrac{1}{\eta} \log \textstyle\sum_{a=1}^{m} \exp(\eta r_a), \quad m = |A|$$

$$\boxed{(\nabla F_\eta(\boldsymbol{r}))_a = \frac{\exp(\eta r_a)}{\sum_b \exp(\eta r_b)}} = x_a$$

Infinitesimal updates stay on one level set

$$\dot{\boldsymbol{r}} = \boldsymbol{g} - \langle \boldsymbol{g}, \boldsymbol{x} \rangle \mathbf{1}$$

$$\frac{d}{dt} F_\eta(\boldsymbol{r}(t)) = \langle \boldsymbol{x}, \dot{\boldsymbol{r}} \rangle = 0$$

$$\max_a r_a(t) \le F_\eta(\boldsymbol{r}(t)) = F_\eta(\mathbf{0}) = \frac{\log m}{\eta}$$

Discrete time: the same potential, plus discretization error

$$\mathrm{Reg}^{(T)} \le \frac{\log m}{\eta} + \eta \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_\infty^2$$

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Quite natural: use the past to inform what to do next.

FTL · RM · MWU · (Generally: FTRL)

2. Improve locally

Take a local improvement (gradient) step.

OGD · MWU · (Generally: OMD)

↑ We continue here

- Online projected gradient ascent

The second philosophy: take a utility-gradient step (and project back)

Projected gradient ascent:

$$\boldsymbol{x}^{(t+1)} := \Pi_{\mathcal{X}}\left(\boldsymbol{x}^{(t)} + \eta \boldsymbol{g}^{(t)}\right)$$

where

$$\Pi_{\mathcal{X}}(\boldsymbol{z}) := \arg\min_{\boldsymbol{x} \in \mathcal{X}} \|\boldsymbol{x} - \boldsymbol{z}\|_2^2$$

OGD: regret bound

$$\boxed{\mathrm{Reg}^{(T)} \leq \frac{D^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_2^2}$$

where $D = \mathrm{diam}_2(\mathcal{X})$. Proof idea: again a telescopic argument (we will see it in the next slide).

On the simplex

$$D \leq \sqrt{2}, \quad \eta = \sqrt{\frac{2}{T|A|}} \quad \Longrightarrow \quad \mathrm{Reg}^{(T)} \leq \sqrt{2T|A|}$$

The OGD proof

2. Now, the term on the right-hand side satisfies

$$\|\hat{\boldsymbol{x}} - (\boldsymbol{x}^{(t)} + \eta \boldsymbol{g}^{(t)})\|_2^2 = \|\hat{\boldsymbol{x}} - \boldsymbol{x}^{(t)}\|_2^2 - 2\eta \langle \boldsymbol{g}^{(t)}, \hat{\boldsymbol{x}} - \boldsymbol{x}^{(t)} \rangle + \eta^2 \|\boldsymbol{g}^{(t)}\|_2^2.$$

3. Rearranging:

$$\langle \boldsymbol{g}^{(t)}, \hat{\boldsymbol{x}} - \boldsymbol{x}^{(t)} \rangle \leq \frac{\|\hat{\boldsymbol{x}} - \boldsymbol{x}^{(t)}\|_2^2 - \|\hat{\boldsymbol{x}} - \boldsymbol{x}^{(t+1)}\|_2^2}{2\eta} + \frac{\eta}{2} \|\boldsymbol{g}^{(t)}\|_2^2$$

4. Sum over time. The distances telescope.

⇒ OGD regret bound

$$\mathrm{Reg}^{(T)} \leq \frac{D^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_2^2$$

where $D = \mathrm{diam}_2(\mathcal{X})$

The guarantees

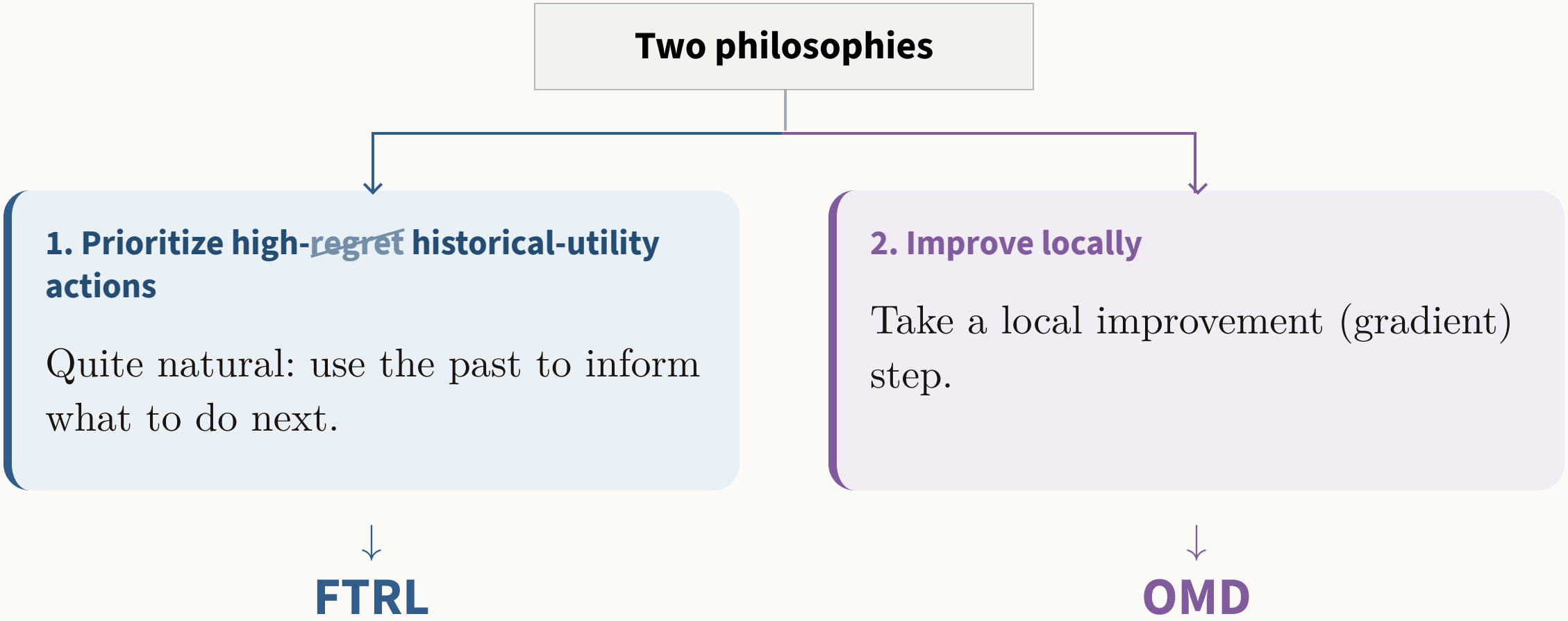
(Assuming $\|\boldsymbol{g}^{(t)}\|_\infty \leq 1$)

Algorithm	Stepsize	Regret bound
FTL	—	$T/2$ on our sequence
RM	None ✨	$2\sqrt{T\|A\|}$
MWU	$\sqrt{\log\|A\|/T}$	$2\sqrt{T\log\|A\|}$
OGD	$\sqrt{2/(T\|A\|)}$	$\sqrt{2T\|A\|}$

Beyond normal-form games

The *big guns*: FTRL and OMD

Two philosophies, two frameworks



Convex strategy sets, concave utilities. Use $\boldsymbol{g}^{(t)} = \nabla u^{(t)}(\boldsymbol{x}^{(t)})$.

Both algorithms work in a tunable geometry

Tune FTRL and OMD to the strategy set

Choose a strongly convex regularizer ψ, also called a *distance-generating function*.

Strongly curved upward

For twice-differentiable ψ, in Euclidean geometry:

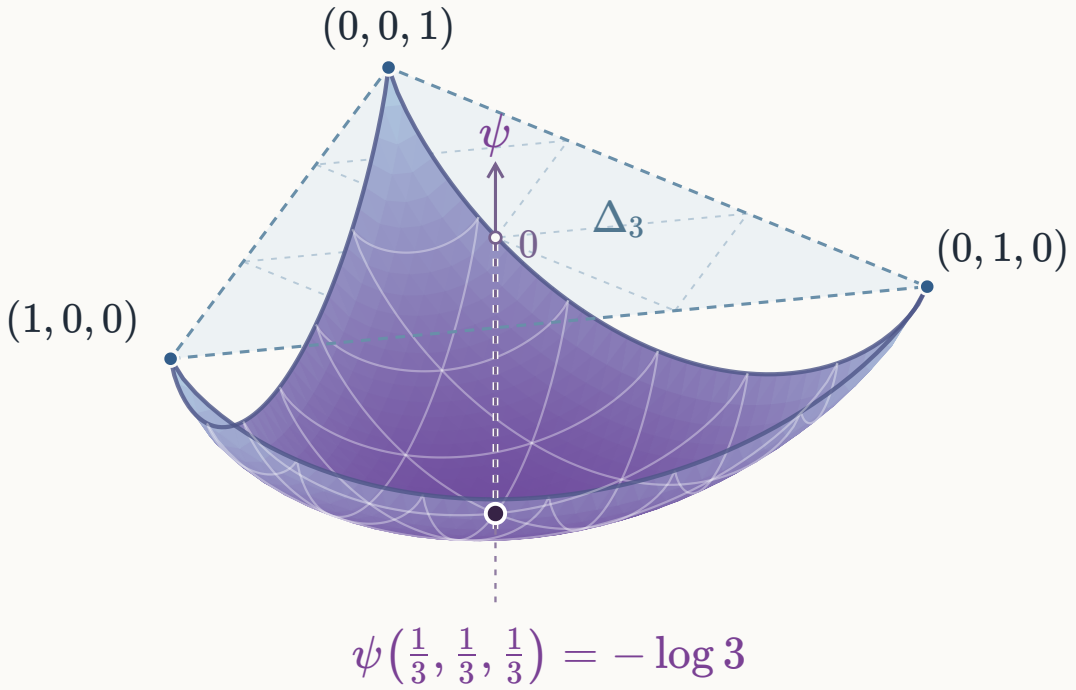
$$\nabla^2 \psi(\boldsymbol{x}) \succeq I$$

A safe choice is always the *squared Euclidean norm*, though it might not be the optimal choice depending on the geometry of the strategy set...

Negative entropy on the simplex

$$\psi(\boldsymbol{x}) = \sum_{a=1}^{3} x_a \log x_a$$

$$x_3 = 1 - x_1 - x_2$$



- Follow-the-regularized-leader

Stabilize the leader with regularization

Balance cumulative utility against a fixed regularizer.

$$\boldsymbol{x}^{(t+1)} = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \left\{ \underbrace{\left\langle \sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)}, \boldsymbol{x} \right\rangle}_{\text{cumulative utility}} - \frac{1}{\eta} \underbrace{\psi(\boldsymbol{x})}_{\text{fixed regularizer}} \right\}$$

Small η

Regularization dominates.
No movement (stay at minimizer of ψ).

Large η

Cumulative utility dominates.
Too much movement ($\equiv$ follow-the-leader).

MWU = FTRL with Entropy

$$\boldsymbol{x}^{(t+1)} = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left\langle \sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)}, \boldsymbol{x} \right\rangle - \frac{1}{\eta}\psi(\boldsymbol{x}) \right\}, \qquad \boxed{\psi(\boldsymbol{x}) = \sum_a x_a \log x_a}.$$

What is the solution to the optimization problem?

Use Lagrange multipliers on the plane of the simplex (ignore nonnegativity constraints):

$$\left(\sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)} \right) - \frac{1}{\eta}(\mathbf{1} + \log \boldsymbol{x}) = \lambda \mathbf{1}$$

$$\implies \log \boldsymbol{x} = c\mathbf{1} + \eta \sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)} \implies \boldsymbol{x} \propto \exp\left\{ \eta \sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)} \right\}$$

which is MWU.

FTRL: Regret Bound

FTRL guarantees

$$\mathrm{Reg}^{(T)} \leq \frac{B}{\eta} + \eta \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_*^2$$

Where $B_{\mathrm{FTRL}} = \max_{\mathcal{X}} \psi - \min_{\mathcal{X}} \psi$

Hence, assuming

$$\|\boldsymbol{g}^{(t)}\|_* \leq G, \quad \eta = \sqrt{\frac{B}{TG^2}} \quad \Longrightarrow \quad \mathrm{Reg}^{(T)} \leq 2G\sqrt{BT}$$

- Online mirror descent

Choose a geometry for each local step

Balance the latest utility feedback against moving from the current strategy.

$$\boldsymbol{x}^{(t+1)} = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \left\{ \underbrace{\langle \boldsymbol{g}^{(t)}, \boldsymbol{x}\rangle}_{\text{latest feedback}} - \frac{1}{\eta} \underbrace{D_\psi(\boldsymbol{x}\|\boldsymbol{x}^{(t)})}_{\text{movement penalty}} \right\}$$

where

$$D_\psi(\boldsymbol{z}\|\boldsymbol{x}) := \psi(\boldsymbol{z}) - \psi(\boldsymbol{x}) - \langle \nabla\psi(\boldsymbol{x}), \boldsymbol{z} - \boldsymbol{x}\rangle$$

Small η

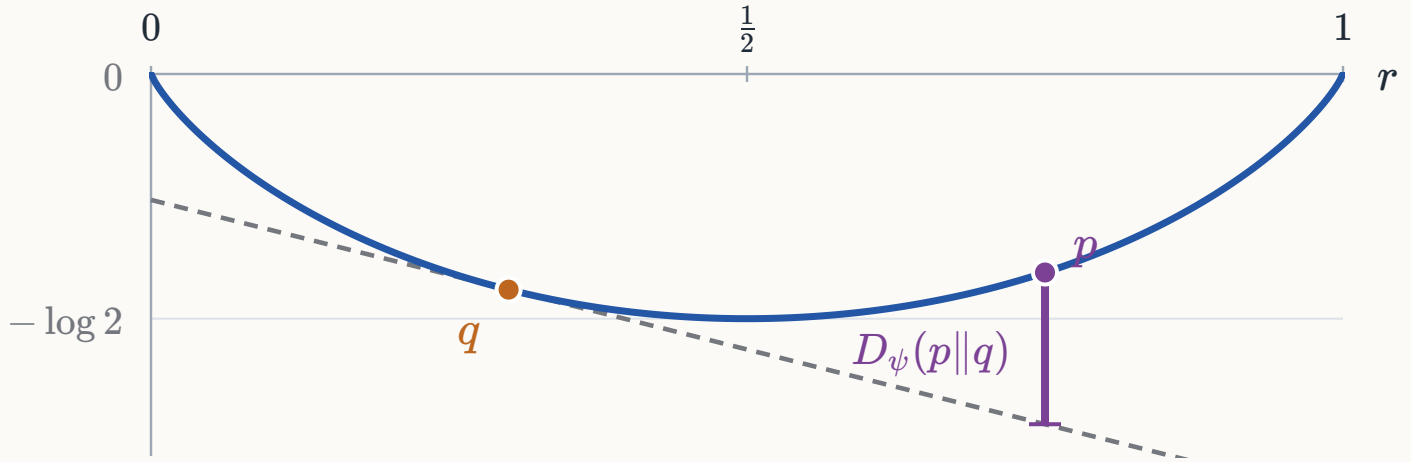
Movement penalty dominates.
Stay near $\boldsymbol{x}^{(t)}$.

Large η

Latest feedback dominates.
Maximize the current linear utility.

Negative entropy and its tangent

$$\psi(\boldsymbol{x}) = \sum_a x_a \log x_a, \qquad \boldsymbol{x} = (p, 1-p), \quad \boldsymbol{q} = (q, 1-q)$$



Drag the points along the curve

Negative entropy

$\psi(r) = r \log r + (1-r)\log(1-r)$

Tangent at current q

Vertical gap = KL divergence

$D_\psi(p\|q) = \mathrm{KL}((p,1-p)\|(q,1-q))$

$D_\psi(p\|q) = 0.43$

Bregman divergence is the height above the tangent

$$D_\psi(\boldsymbol{x}\|\boldsymbol{q}) = \sum_a x_a \log(x_a/q_a) = D_{\mathrm{KL}}(\boldsymbol{x}\|\boldsymbol{q}).$$

The geometry chooses the algorithm

Regularizer ψ	Divergence	Algorithm
$\sum_a x_a \log x_a$	D_{KL}	MWU FTRL and OMD
$\frac{1}{2}\|\boldsymbol{x}\|_2^2$	$\frac{1}{2}\|\boldsymbol{z}-\boldsymbol{x}\|_2^2$	OGD OMD

MWU and OGD are instances of general principles

Entropy gives MWU. Euclidean mirror descent gives OGD; quadratic FTRL can differ under constraints.

OMD: Regret Bound

OMD guarantees

$$\mathrm{Reg}^{(T)} \leq \frac{B}{\eta} + \eta \sum_{t=1}^{T} \|\boldsymbol{g}^{(t)}\|_*^2$$

Where

$$B_{\mathrm{OMD}} = \max_{\boldsymbol{x} \in \mathcal{X}} D_\psi(\boldsymbol{x} \| \boldsymbol{x}^{(1)})$$

Hence, assuming

$$\|\boldsymbol{g}^{(t)}\|_* \leq G, \quad \eta = \sqrt{\frac{B}{TG^2}} \quad \Longrightarrow \quad \mathrm{Reg}^{(T)} \leq 2G\sqrt{BT}$$

FTRL and OMD can overlap

For example: OMD with negative entropy

$$\psi(\boldsymbol{x}) = \sum_a x_a \log x_a, \qquad D_\psi(\boldsymbol{x} \| \boldsymbol{x}^{(t)}) = \sum_a x_a \log(x_a / x_a^{(t)}).$$

gives rise to

$$\log x_a^{(t+1)} = \log x_a^{(t)} + \eta g_a^{(t)} + c.$$

And hence

$$x_a^{(t+1)} = \frac{x_a^{(t)} \exp(\eta g_a^{(t)})}{\sum_b x_b^{(t)} \exp(\eta g_b^{(t)})}.$$

Unrolling gives the same update as **entropic FTRL**:

$$\boxed{x_a^{(t+1)} \propto \exp\left(\eta \sum_{\tau=1}^{t} g_a^{(\tau)}\right).}$$

Appendix

Faster rates & Iterate convergence

Optimism: add a momentum term

Nonstationary need not mean adversarial

Use the latest gradient to anticipate the next one.

In reflected OMD, replace the feedback by

$$\widehat{\boldsymbol{g}}^{(t)} = \boldsymbol{g}^{(t)} + \underbrace{\left(\boldsymbol{g}^{(t)} - \boldsymbol{g}^{(t-1)}\right)}_{\text{momentum correction}} = 2\boldsymbol{g}^{(t)} - \boldsymbol{g}^{(t-1)}$$

In zero-sum self-play, this can damp cycling: a negative-momentum effect.

Optimistic FTRL / OMD in zero-sum self-play

$\gamma(\overline{\boldsymbol{x}}, \overline{\boldsymbol{y}}) = O(1/T)$ with suitable variants and a sufficiently small fixed stepsize.

Optimism in both frameworks

Add a momentum correction. Set $\boldsymbol{g}^{(0)} = \mathbf{0}$.

FTRL: count the latest gradient once more

$$\boldsymbol{x}^{(t+1)} = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left\langle \sum_{\tau=1}^{t} \boldsymbol{g}^{(\tau)} + \boldsymbol{g}^{(t)}, \boldsymbol{x} \right\rangle - \frac{1}{\eta}\psi(\boldsymbol{x}) \right\}$$

Reflected OMD: add the change in gradients

$$\boldsymbol{x}^{(t+1)} = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left\langle \boldsymbol{g}^{(t)} + \left(\boldsymbol{g}^{(t)} - \boldsymbol{g}^{(t-1)}\right), \boldsymbol{x} \right\rangle - \frac{1}{\eta} D_{\psi}(\boldsymbol{x} \| \boldsymbol{x}^{(t)}) \right\}$$

The frameworks can differ under constraints. Further variants: Supplementary S04.

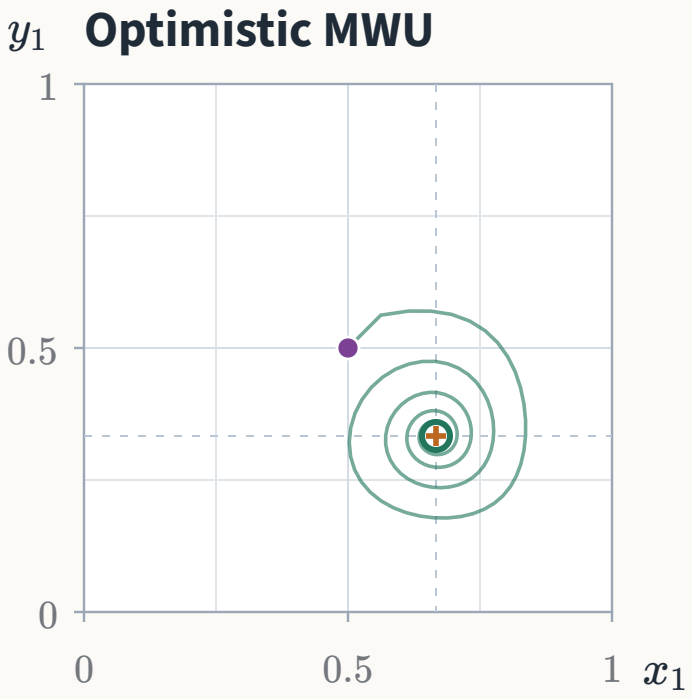
Faster regret in general games

Method / setting	Each player's regret	CCE error
Optimistic FTRL / OMD Syrgkanis et al. (2015)	$O(T^{1/4})$	$O(T^{-3/4})$
Optimistic MWU · normal-form games Daskalakis, Fishelson, Golowich (2021)	$O(\log^4 T)$	$O(\log^4 T/T)$
Specialized optimistic FTRL · convex games Farina et al. (2022) · lifted space, log regularizer	$O(\log T)$	$O(\log T/T)$

Cautious optimism: $O_T(\log T)$ — Soleymani, Piliouras & Farina (2025).
Constant $O_T(1)$ regret (2026, normal-form): Liu et al.; Abbadi et al.; Soleymani & Piliouras.

Fast regret gives fast convergence of empirical play
$\mathrm{Reg}_i^{(T)} \le R(T)\ \forall i \quad \Longrightarrow \quad \varepsilon_T \le R(T)/T$. What about the last iterate?

Optimistic learning dynamics

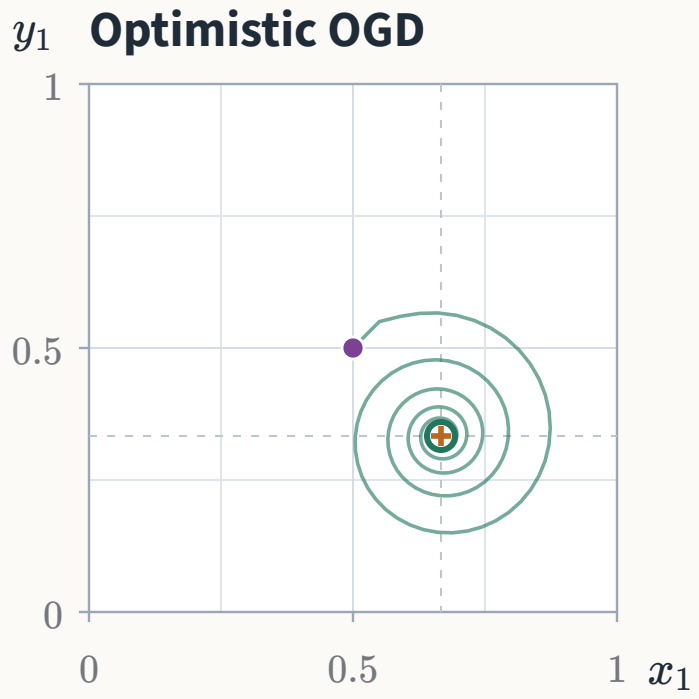


$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

- Start
- Last iterate
- Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap

2.7×10^{-7}



$\eta_{\mathrm{MWU}} = 0.25$

$\eta_{\mathrm{OGD}} = 0.1$

- Start
- Last iterate
- Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap

3.0×10^{-6}

(See Supplementary S04.)

Optimistic MWU can converge slowly

Ordinary MWU can cycle. Does optimism always fix the problem?

$$U_\delta = \begin{pmatrix} \frac{1}{2} - \delta & \frac{1}{2} \\ 1 & 0 \end{pmatrix}, \quad u_1 = \boldsymbol{x}^\mathsf{T} U_\delta \boldsymbol{y}, \quad u_2 = -u_1, \quad 0 < \delta < \tfrac{1}{2}$$

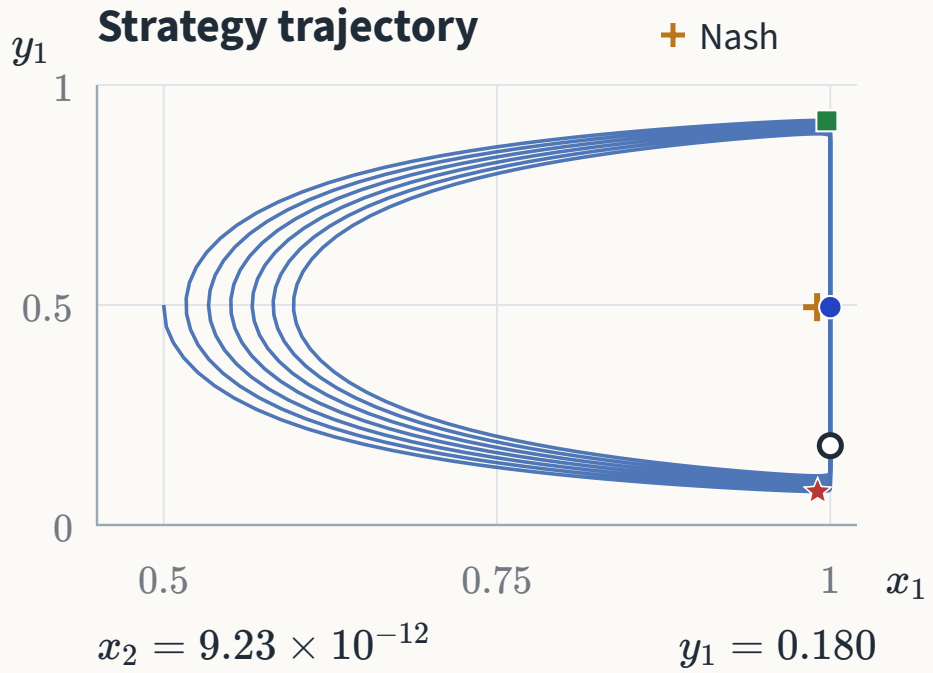
$$\boldsymbol{x}^* = \left(\frac{1}{1+\delta}, \frac{\delta}{1+\delta} \right), \qquad \boldsymbol{y}^* = \left(\frac{1}{2(1+\delta)}, \frac{1+2\delta}{2(1+\delta)} \right)$$

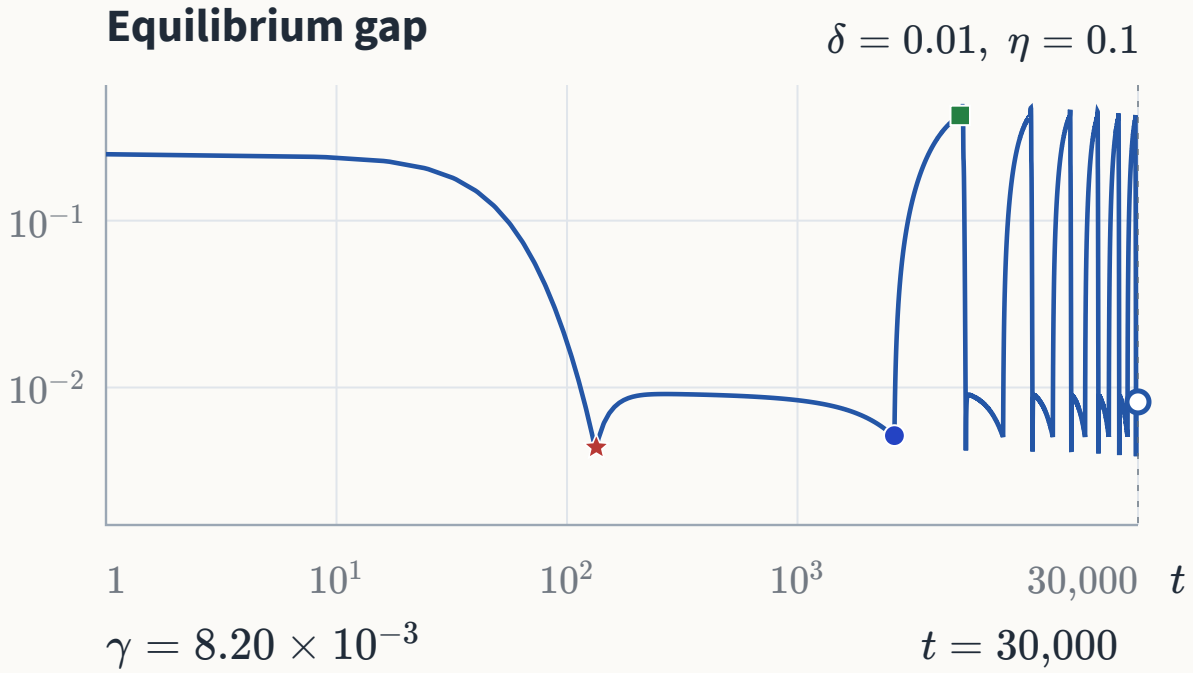
An equilibrium close to the boundary

Optimistic MWU can retain an overwhelming preference for an action and overshoot the equilibrium.

→ Cai, Farina, Grand-Clément, Kroer, Lee, Luo, Zheng (2024).

OMWU: trajectory and equilibrium gap





The limit of optimistic MWU

No uniform last-iterate convergence rate

For fixed-stepsize OMWU, no function of action counts and time alone gives a vanishing bound for every bounded-payoff game.

$$\gamma(\boldsymbol{x}^{(T)}, \boldsymbol{y}^{(T)}) \leq f(|A_1|, |A_2|, T), \qquad f(|A_1|, |A_2|, T) \longrightarrow 0$$

Already impossible uniformly over 2×2 games with payoffs in $[0, 1]$.

Convergence in a fixed game is still possible

The rate must depend on the game. Optimistic OGD, by contrast, has a uniform $O(1/\sqrt{T})$ last-iterate guarantee.

Cai et al., Fast Last-Iterate Convergence of Learning in Games Requires Forgetful Algorithms.